MATH 241: Calculus III

Fall 2026

Lecture 24: Curl and Divergence

Background & Motivation

Green's Theorem involved the quantity $Q_x - P_y$. Today we give it a name: the **curl**. We also introduce the **divergence**, which measures expansion or compression. These two differential operators are the vocabulary of the final theorems of the course (Stokes' and the Divergence Theorem).

1. Curl: Measuring Rotation

Definition 1: Curl of a Vector Field

For $\mathbf{F} = \langle P, Q, R\rangle$, the **curl** is (all the same thing, different notation):

$$\operatorname{curl}\mathbf{F} = \boldsymbol{\nabla}\times\mathbf{F} = \nabla\times\mathbf{F} = \langle R_y - Q_z,\ \ P_z - R_x,\ \ Q_x - P_y\rangle$$

How to compute (component by component):

- $\hat{\mathbf{i}}$: $\dfrac{\partial R}{\partial y} - \dfrac{\partial Q}{\partial z}$ $\quad\hat{\mathbf{j}}$: $\dfrac{\partial P}{\partial z} - \dfrac{\partial R}{\partial x}$ $\quad\hat{\mathbf{k}}$: $\dfrac{\partial Q}{\partial x} - \dfrac{\partial P}{\partial y}$

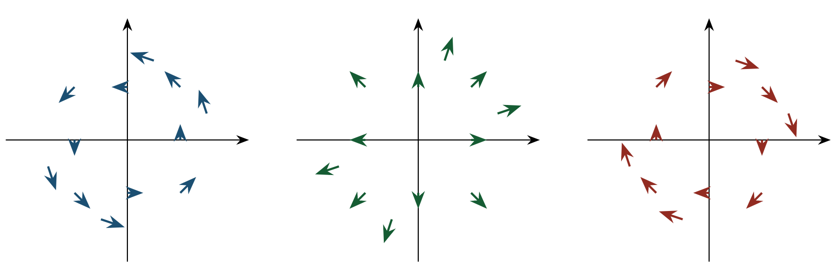
Memory trick: each component uses the two variables it doesn't point along, in cyclic order ($x \to y \to z \to x \to \cdots$). Always subtract in "reverse" order. If you know determinants, this equals $\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial_x & \partial_y & \partial_z \\ P & Q & R \end{vmatrix}$, but that is not required.

– **Interpretation:** measures the local rotation ("swirling") of the field at a point

– $\boldsymbol{\nabla}\times\mathbf{F} = \mathbf{0} \implies$ **irrotational** (no spinning)

– Direction of $\boldsymbol{\nabla}\times\mathbf{F}$ = axis of rotation (right-hand rule), magnitude = rate of rotation

Visualizing Curl



Positive curl $\langle -y, x\rangle$, curl $= 2$ | **Zero curl** $\langle x, y\rangle$, curl $= 0$ | **Negative curl** $\langle y, -x\rangle$, curl $= -2$

The Paddlewheel Test

Think of placing a tiny paddlewheel at a point in the flow. The fluid pushes on each blade. If the flow is faster or differently directed on one side than the other, the imbalance creates a torque and the wheel spins.

- **Positive curl** ($\nabla \times \mathbf{F} > 0$ in 2D): paddlewheel spins **CCW**. Right-hand rule: thumb out of page ($+\hat{\mathbf{k}}$).
- **Negative curl** ($\nabla \times \mathbf{F} < 0$ in 2D): paddlewheel spins **CW**. Right-hand rule: thumb into page ($-\hat{\mathbf{k}}$).
- **Zero curl**: paddlewheel doesn't spin, even if fluid is moving fast.

Key insight: curl measures *local spin*, not global shape. A field can have curved streamlines but zero curl, and *straight* streamlines but nonzero curl!

Example — shear flow: In $\mathbf{F} = \langle y, 0 \rangle$ every arrow is horizontal, yet the paddlewheel spins. The top blade ($y > 0$) is pushed right; the bottom blade ($y < 0$) is pushed left. This imbalance $\Longrightarrow$ **CW** rotation, and indeed $Q_x - P_y = 0 - 1 = -1 < 0$. Negative curl despite no circular streamlines.

In 3D, $\nabla \times \mathbf{F}$ is a vector: its **direction** is the axis of fastest rotation (right-hand rule) and its **magnitude** is the rate of rotation.

Example 1: Determining Curl Sign from the Paddlewheel

For each field, predict the paddlewheel behavior at the origin, then verify by computing the 2D curl.

(a) $\mathbf{F} = \langle 0,\ x \rangle$
(b) $\mathbf{F} = \langle -y,\ 0 \rangle$
(c) $\mathbf{F} = \langle y,\ x \rangle$

(a) Right of origin ($x > 0$): upward push. Left ($x < 0$): downward push. $\Longrightarrow$ **CCW**.
$Q_x - P_y = 1 - 0 = 1 > 0$ ✓ Positive curl (CCW).

(b) Above origin ($y > 0$): leftward push. Below ($y < 0$): rightward push. $\Longrightarrow$ **CCW**.
$Q_x - P_y = 0 - (-1) = 1 > 0$ ✓ Positive curl (CCW). Different-looking field, same curl as (a)!

(c) From $P = y$: top-right, bottom-left (CW tendency). From $Q = x$: right-up, left-down (CCW tendency). They **cancel**.
$Q_x - P_y = 1 - 1 = 0$ ✓ Zero curl. In fact $\mathbf{F} = \nabla(xy)$ — conservative.

2D Reduction

For $\mathbf{F} = \langle P, Q \rangle$ (2D, so $R = 0$), only the $\hat{\mathbf{k}}$ component survives: $\nabla \times \mathbf{F} = (Q_x - P_y)\,\hat{\mathbf{k}}$. We call the scalar $Q_x - P_y$ the **2D curl** — exactly the integrand in Green's Theorem!

Example 2: 3D Curl Computation

Find $\nabla \times \mathbf{F}$ for $\mathbf{F} = \left\langle xz,\ xyz,\ -y^2 \right\rangle$.
$P = xz, \quad Q = xyz, \quad R = -y^2.$
$\hat{\mathbf{i}}$: $R_y - Q_z = (-y^2)_y - (xyz)_z = -2y - xy$
$\hat{\mathbf{j}}$: $P_z - R_x = (xz)_z - (-y^2)_x = x - 0 = x$
$\hat{\mathbf{k}}$: $Q_x - P_y = (xyz)_x - (xz)_y = yz - 0 = yz$
$\nabla \times \mathbf{F} = \boxed{\langle -2y - xy,\ \ x,\ \ yz \rangle}$

Example 3: 2D Curl: Rotation vs. Source

Compute the 2D curl of (a) $\mathbf{F} = \langle -y, x\rangle$ and (b) $\mathbf{F} = \langle x, y\rangle$.
(a) $Q_x - P_y = (x)_x - (-y)_y = 1 - (-1) = 2$. Positive $\implies$ CCW rotation everywhere.
(b) $Q_x - P_y = (y)_x - (x)_y = 0 - 0 = 0$. Irrotational! Despite pointing outward everywhere, there is no local spinning. (This field is conservative: $\mathbf{F} = \nabla\left(\frac{x^2+y^2}{2}\right)$.)

2. Divergence: Measuring Expansion

Definition 2: Divergence of a Vector Field

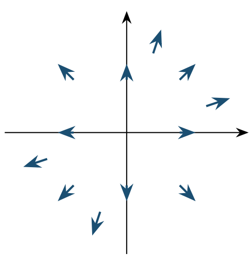
For $\mathbf{F} = \langle P, Q, R\rangle$, the **divergence** is (same thing, different notation):

$$\operatorname{div}\mathbf{F} = \operatorname{div}\mathbf{F} = \nabla\cdot\mathbf{F} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

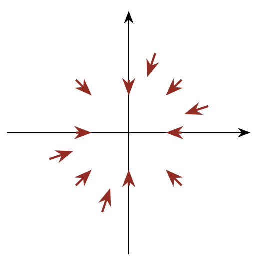
Just add up how much each component changes along its own direction.

– **Interpretation:** measures the net **"outflow" or expansion at a point**
– $\operatorname{div}\mathbf{F} > 0$: **source** (fluid spreading out) $\operatorname{div}\mathbf{F} < 0$: **sink** (fluid compressing) $\operatorname{div}\mathbf{F} = 0$: **incompressible**

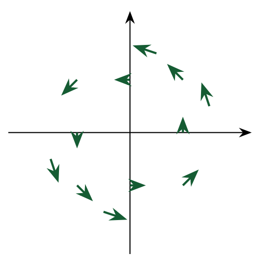
Visualizing Divergence



Source (div > 0)
$\langle x, y\rangle$, div $= 2$



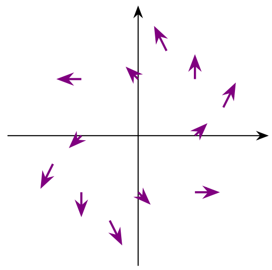
Sink (div < 0)
$\langle -x, -y\rangle$, div $= -2$



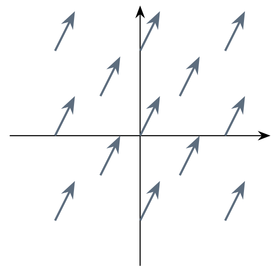
Incompressible (div $= 0$)
$\langle -y, x\rangle$, div $= 0$

Both Properties / Neither

Curl and divergence are **independent**. A field can have any combination:



Both (curl $\neq 0$, div $\neq 0$)
$\langle x-y,\ x+y\rangle$
curl $= 2$, div $= 2$



Neither (curl $= 0$, div $= 0$)
$\langle 1,\ 2\rangle$
Uniform flow

Example 4: Incompressible Pipe Flow

A velocity field in a pipe is $\mathbf{F} = \langle 2x, -y, -z\rangle$. Is the flow incompressible?
$\operatorname{div}\mathbf{F} = (2x)_x + (-y)_y + (-z)_z = 2 + (-1) + (-1) = 0$
Yes — $\operatorname{div}\mathbf{F} = 0$, so the flow is **incompressible**. The fluid speeds up along x (pipe narrows) but compensates by slowing in y and z. No fluid is created or destroyed.

Example 5: Classifying a Field

For $\mathbf{F} = \langle x, y, z\rangle$, find $\operatorname{div}\mathbf{F}$ and $\nabla \times \mathbf{F}$. Interpret.
$\operatorname{div}\mathbf{F} = 1 + 1 + 1 = 3$ (source everywhere — uniform expansion)
$\nabla \times \mathbf{F}$: $\hat{\mathbf{i}}$: $0 - 0 = 0$, $\hat{\mathbf{j}}$: $0 - 0 = 0$, $\hat{\mathbf{k}}$: $0 - 0 = 0$. So $\nabla \times \mathbf{F} = \mathbf{0}$ (irrotational).
Like an explosion: everything moves outward, nothing rotates.

Example 6: Another Classification

For $\mathbf{F} = \langle -y, x, 0\rangle$, find $\operatorname{div}\mathbf{F}$ and $\nabla \times \mathbf{F}$. Interpret.
$\operatorname{div}\mathbf{F} = (-y)_x + (x)_y + (0)_z = 0$ (incompressible)
$\nabla \times \mathbf{F}$: $\hat{\mathbf{i}}$: $0 - 0 = 0$, $\hat{\mathbf{j}}$: $0 - 0 = 0$, $\hat{\mathbf{k}}$: $1 - (-1) = 2$. So $\nabla \times \mathbf{F} = 2\hat{\mathbf{k}}$ (CCW about z).
Like a merry-go-round: everything spins, nothing expands or compresses.

3. Key Identities

Vector Calculus Identities

1. $\nabla \times (\nabla f) = \mathbf{0}$ — conservative fields are always irrotational (by Clairaut's: $f_{xy} = f_{yx}$)
   Contrapositive: if $\nabla \times \mathbf{F} \neq \mathbf{0}$, then $\mathbf{F}$ is **not** conservative.
2. $\operatorname{div}(\nabla \times \mathbf{F}) = 0$ — curl fields are always divergence-free (Clairaut's again)
3. **Laplacian:** $\nabla^2 f = \operatorname{div}(\nabla f) = f_{xx} + f_{yy} + f_{zz}$ — divergence of the gradient
4. **Product rules:**
   $\nabla(fg) = f\nabla g + g\nabla f$ $\qquad \operatorname{div}(f\mathbf{F}) = f\ \operatorname{div}\mathbf{F} + \mathbf{F}\cdot\nabla f$ $\qquad \nabla \times (f\mathbf{F}) = f\,\nabla \times \mathbf{F} + \nabla f \times \mathbf{F}$

Operator Relationships

$$\text{Scalar field } f \xrightarrow{\text{grad}} \text{Vector field } \mathbf{F} \xrightarrow{\text{curl}} \text{Vector field } \mathbf{G} \xrightarrow{\text{div}} \text{Scalar field } h$$

The two zero identities say: grad $\to$ curl $= \mathbf{0}$, and curl $\to$ div $= 0$. In other words, the "derivative of a derivative" is zero in this chain.

Example 7: Verifying $\boldsymbol{\nabla} \times (\nabla f) = \mathbf{0}$

Let $f(x, y, z) = x^2yz + \sin(yz)$. Verify $\boldsymbol{\nabla} \times (\nabla f) = \mathbf{0}$.
Gradient: $\nabla f = \big\langle 2xyz,\ x^2z + z\cos(yz),\ x^2y + y\cos(yz)\big\rangle$
Curl:
$\hat{\mathbf{i}}$: $(x^2y+y\cos(yz))_y-(x^2z+z\cos(yz))_z = [x^2+\cos(yz)-yz\sin(yz)]-[x^2+\cos(yz)-yz\sin(yz)] = 0$ ✓
$\hat{\mathbf{j}}$: $(2xyz)_z - (x^2y + y\cos(yz))_x = 2xy - 2xy = 0$ ✓
$\hat{\mathbf{k}}$: $(x^2z + z\cos(yz))_x - (2xyz)_y = 2xz - 2xz = 0$ ✓
This always works by Clairaut's theorem ($f_{xy} = f_{yx}$, etc.).

Example 8: Proving $\operatorname{div}(\boldsymbol{\nabla} \times \mathbf{F}) = 0$

Show that $\operatorname{div}(\boldsymbol{\nabla} \times \mathbf{F}) = 0$ for any smooth $\mathbf{F} = \langle P, Q, R\rangle$.
$\boldsymbol{\nabla} \times \mathbf{F} = \langle R_y - Q_z,\ P_z - R_x,\ Q_x - P_y\rangle$
$\operatorname{div}(\boldsymbol{\nabla} \times \mathbf{F}) = (R_y - Q_z)_x + (P_z - R_x)_y + (Q_x - P_y)_z$
$= R_{yx} - Q_{zx} + P_{zy} - R_{xy} + Q_{xz} - P_{yz}$
By Clairaut's: $R_{yx} = R_{xy}$, $Q_{zx} = Q_{xz}$, $P_{zy} = P_{yz}$. Every term cancels. □

4. Green's Theorem Reformulated

Green's Theorem via Curl

$$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \iint_D (\boldsymbol{\nabla} \times \mathbf{F}) \cdot \hat{\mathbf{k}}\, dA$$

The circulation of $\mathbf{F}$ around C equals the total curl over D. This is the 2D special case of **Stokes' Theorem** (coming soon!).

Practice Problems

Problem 1. Compute $\operatorname{div}\mathbf{F}$ and $\boldsymbol{\nabla} \times \mathbf{F}$ for $\mathbf{F} = \big\langle x^2,\, y^2,\, z^2\big\rangle$.

Answer: $\operatorname{div}\mathbf{F} = 2x + 2y + 2z, \quad \boldsymbol{\nabla} \times \mathbf{F} = \mathbf{0}$

Problem 2. Show that $\mathbf{F} = \langle yz,\, xz,\, xy\rangle$ is both irrotational and incompressible.

Answer: $\boldsymbol{\nabla} \times \mathbf{F} = \mathbf{0}, \quad \operatorname{div}\mathbf{F} = 0$

Problem 3. Find $\nabla \times \mathbf{F}$ and $\operatorname{div} \mathbf{F}$ for $\mathbf{F} = \langle y^2 z,\ x^2 z,\ x^2 y\rangle$.

Answer: $\nabla \times \mathbf{F} = \langle 0,\ y^2 - 2xy,\ 2z(x-y)\rangle, \quad \operatorname{div} \mathbf{F} = 0$

Problem 4. Compute $\nabla \times \mathbf{F}$ for $\mathbf{F} = \langle e^x \cos y,\ -e^x \sin y,\ 0\rangle$. What does the result tell you?

Answer: $\nabla \times \mathbf{F} = \mathbf{0} \quad \Longrightarrow$ conservative, potential $f = e^x \cos y$

Problem 5. Verify $\operatorname{div}(f\mathbf{F}) = f\ \operatorname{div} \mathbf{F} + \mathbf{F} \cdot \nabla f$ for $f = xy$ and $\mathbf{F} = \langle 1, 1, 0\rangle$.

Answer: both sides $= x + y\ \checkmark$

Problem 6. Prove: if $\mathbf{F} = \nabla f$ and f is harmonic ($\nabla^2 f = 0$), then $\mathbf{F}$ is both irrotational and incompressible.

Irrotational: $\nabla \times \mathbf{F} = \nabla \times (\nabla f) = \mathbf{0}$ (identity 1).

Incompressible: $\operatorname{div} \mathbf{F} = \operatorname{div}(\nabla f) = \nabla^2 f = 0$ (since f is harmonic). $\square$

Problem 7. Without computing, predict whether the 2D curl is positive, negative, or zero at the origin. Then verify.

(a) $\mathbf{F} = \langle -2y,\ 3x\rangle$ (b) $\mathbf{F} = \langle y,\ y\rangle$ (c) $\mathbf{F} = \langle x^2,\ -y^2\rangle$

Answer: (a) $Q_x - P_y = 3 - (-2) = 5$ (positive, CCW). (b) $Q_x - P_y = 0 - 1 = -1$ (negative, CW). (c) $Q_x - P_y = 0$ (irrotational).

Problem 8. The velocity field of a whirlpool is modeled by $\mathbf{F} = \langle -y, x, 2z\rangle$. Find $\nabla \times \mathbf{F}$ and $\operatorname{div} \mathbf{F}$. Does this flow rotate, expand, or both?

$\nabla \times \mathbf{F} = \langle 0 - 0,\ 0 - 0,\ 1 - (-1)\rangle = 2\hat{\mathbf{k}}$ (CCW rotation about z-axis)

$\operatorname{div} \mathbf{F} = 0 + 0 + 2 = 2 \quad$ (expanding vertically)

Both! Like a tornado: spinning while air is pulled upward.